MILITARY SPECIFICATION

PROPELLANT, M2, FOR CARTRIDGE, 37MM, TP, M63 Mod. 1

1. SCOPE

1.1 This specification covers one type of propellant designated as M2 for use in Cartridge, 37MM, TP, M63 Mod. 1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-G-155 - Graphite (For Use In Ammunition)
MIL-P-156 - Potassium Nitrate
MIL-N-244 - Nitrocellulose (For Use In Explosive)
MIL-N-246 - Nitroglycerin
JAN-E-255 - Ethyl Centralite (Carbamite)
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-286 - Propellants: Sampling, Inspection and Testing.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

FSC: 1375

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DRAWINGS

76-4-46 - Box, Packing, with Metal Liner, M24 for Smokeless Powder. Assembly and Details.
76-4-56 - Box, Packing, Metal-Wood, M17 for Smokeless Powder.
8858848 - Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for shipment of Propellants.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publication required by contractors in connecton with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-The constituent materials shall comply with the requirements of the applicable specification as follows:

CONSTITUENT MATERIALS

	Conforming to Specification
Nitrocellulose	MIL-N-244 Grade C, Type I
Nitroglycerin	MIL-N-246, Type I
Potassium Nitrate	MIL-P-156, Class 2
Ethyl Centralite	JAN-G-255
Graphite	JAN-G-155 Grade III or IV

3.2 Composition.-The composition, exclusive of total volatiles shall be as specified in Table I when determined as specified in 4.3.1.

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TABLE I

Constituent	Percent
Nitrocellulose	77.45 plus or minus 2.00
Nitroglycerin	19.50 plus or minus 1.00
Potassium Nitrate	2.15 plus or minus 0.25
Ethyl Centralite	0.60 plus or minus 0.15
Graphite 1	0.30 plus or minus 0.10

Exclusive of the added graphite glaze (see 3.3)

3.3 Graphite glaze.The propellant may be glazed in addition to the graphite used in making up the composition.

3.4 Total volatiles.-The total volatiles content shall not exceed 1.87 percent when determined as specified in 4.3.2.

3.5 Moisture.-The moisture content shall not exceed 0.70 percent when determined as specified in 4.3.3.

3.6 Hygroscopicity.-The hygroscopicity shall not exceed 1.50 percent, at 90 percent relative humidity and 30 degree Centigrade, when determined as specified in 4.3.3.

3.7 Stability test, 120 degrees centigrade.-The propellant shall not cause complete fading of the methyl violet test paper to salmon pink in less than 40 minutes and shall not explode in less than 5 hours when determined as specified in 4.3.4.

3.8 Form and dimensions.

3.8.1 Form.-The propellant shall be in the form of cylindrical multi-perforated grains.

3.8.2 Dimension.-The following requirements regarding grain dimensions shall govern unless authorization for departure is given, by the procuring activity concerned, prior to manufacture of the propellant.

3.8.2.1 Length - diameter ratio.-The average grain length (L) shall be from 2.10 to 2.50 times the average grain diameter (D).

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3.8.2.2 Length - diameter uniformity.-The length and diameter of the grain shall comply with the uniformity requirements shown in Table II.

TABLE II

Standard Deviation of Individual Dimensions Expressed as Percentage of Mean Dimension (see 6.3).

Dimension	Maximum
L	6.25
D(grain 0.2 inches or more in diameter)	3.125

3.8.2.3 Grain diameter - perforation diameter ratio.-The average grain diameter (D) shall be not less than 5 nor more than 15 times the average diameter of the perforations (d).

3.8.2.4 Web measurements.-The difference between the average outer web thickness (Wo) and the average inner web thickness (W') shall be not more than 15 percent of the average thickness (Wa).

3.9 Ballistics.-The propellant, when fired in the weapon and with the weight of projectile for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.6.

3.9.1 Velocity and Pressure.-The propellant, at 70 degrees Fahrenheit, shall be assessed to impart a service velocity of 2100 feet per second to the projectile without exceeding a maximum rated pressure of 27,000 pounds per square inch.

3.9.2 Uniformity of velocity.-The standard deviation in velocity for the lot, at 70 degrees Fahrenheit, shall not exceed 6.5 feet per second.

3.10 Workmanship.-The best commercial practices shall be used in the formulation of the propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of Propellant Description Sheet, OO Form-1204, containing the applicable data.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2.1 Lot formation.-A lot shall consist of one or more batches of M2 propellant produced by one manufacturer, in accordance with the same specification and same specification revision and under one continuous set of operating conditions. Each batch shall consist of that quantity of M2 propellant which has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each minor defect.

4.2.2.1 Box or Container, prior to filling (see dwg. 76-4-46, or 76-4-56 as applicable).

Categories	Defects	Method of Inspection	Code No (see 6.5)
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Cracks or holes in metal liner (as applicable)	Visual	01001
102.	Foreign matter on interior	Visual	01002
Minor:	None defined		

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4.2.2.2 Sealed Box or Container (see dwg 76-4-46, or 76-4-56 as applicable)

Categories	Defects	Method of Inspection	Code No
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Cover incompletely closed	Visual-Manual	02001
102.	Gasket missing or broken	Visual	02002
103.	Wooden cover split or otherwise damaged (when applicable)	Visual	02003
Minor:	AQL 2.50 percent		
201.	Car seal missing unsealed or improperly positioned	Visual-Manual	02004
202.	Crack or hole in metal container	Visual	02005
203.	Hardware improperly engaged	Visual	02006
204.	Marking misleading or unidentifiable	Visual	02007
205.	Bare areas in exterior coating of metal container the sum of which is in excess of 1/4 square inch	Visual	02008

4.2.3 Testing

4.2.3.1 Chemical composition physical test and surveillance.- Six pounds of propellant shall be selected at random from the lot. One pound shall be used for chemical composition and physical tests. If the sample fails to comply with the requirements specified the lot shall be rejected.

a.	Chemical Composition	
1.	Nitrocellulose	03001
2.	Nitroglycerin	03002
3.	Potassium nitrate	03003
4.	Ethyl Centralite	03004
5.	Graphite	03005
b.	Total volatiles	04001
c.	Moisture	05001
d.	Hygroscopicity	06001
e.	Stability	07001

The remaining five pounds shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey for 65.5 degrees Centigrade surveillance test. All samples shall be packed in air-tight containers and shall be marked to show the propellant designation, the projectile or cartridge for which manufactured, date of sampling, contract number and number of pounds in the lot. (see 6.2)

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4.2.3.2 Form and dimensions (see 3.8) Major defect.- Thirty grains shall be selected from each lot for these examinations. If the sample fails to meet the requirements specified the lot shall be rejected.

		Code Number
a.	Form	08001
b.	Length-diameter uniformity	08002
c.	Wet measurements	08003

4.2.3.3 Ballistics-(see 3.9) Sufficient propellant shall be randomly selected from the lot to provide for charge establishment, uniformity and informational firings and to provide for a possible ten percent loss due to incomplete measurements. A ten-round uniformity series shall be fired after conditioning the rounds to 70 degrees Fahrenheit.

4.2.3.3.1 Velocity and pressure (critical defect) Code No. 09001.-The lot shall be rejected if the calculated pressure at the charge recommended for prescribed service velocity exceeds the average maximum pressure limitation specified in 3.9.1.

4.2.3.3.2 Velocity uniformity.-(Major defect) Code No. 09002. The lot shall be rejected if the standard deviation of the uniformity series times the factor 0.73 fails to comply with the applicable requirement (see 3.9.2).

4.2.3.4 Retest.-If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional charges as required.

4.2.3.5 Air pressure test(Major defect) Code No. 01001.- The containers shall be tested 100 percent in accordance with 4.3.7. If any container fails to meet the requirements it shall be rejected and removed from the lot (see dwg. 8858848).

4.3 Test method and procedures.

4.3.1 Composition

4.3.1.1 Nitroglycerin.-The percentage of nitroglycerin shall be determined in accordance with Standard MIL-STD-286, Method 208.1 using pentane-methylene chloride azeotrope (two volumes of technical grade pentane to one volume of methylene chloride) as the solvent for extraction, except that the nitroglycerin content shall be calculated on a graphite (glaze) and total volatiles - free basis.

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4.3.1.2 Ethyl centralite.-The percentage of ethyl centralite shall be determined as specified in Standard MIL-STD-286 method 202.2 using - methylene chloride azeotrope as the solvent for extraction, except that the ethyl centralite content shall be calculated on a graphite (glaze) and total volatiles free basis.

4.3.1.3 Potassium Nitrate.-The percentage of potassium nitrate shall be determined in accordance with Standard MIL-STD-286 method 310.4 except that the potassium nitrate shall be calculated on a graphite (glaze) and total volatiles free basis.

4.3.1.4 Graphite.-The percentage of graphite shall be determined in accordance with Standard MIL-STD-286 method 308.1.

4.3.1.5 Nitrocellulose.-The percentage of nitrocellulose shall be determined in accordance with Standard MIL-STD-286, method 209.2.

4.3.2 Total volatiles.-The percentage of total volatiles shall be determined in accordance with Standard MIL-STD-286, method 103.3.

4.3.3 Moisture.-The percentage of moisture shall be determined in accordance with Standard MIL-STD-286, method 102.1.

4.3.4 Stability test, 120 degrees Centigrade.-The heat test shall be conducted in accordance with Standard MIL-STD-286, method 404.1.

4.3.5 Form and dimensions of grains.-Examination for form shall be visual; measurements and calculation for dimensions shall be conducted in accordance with Standard MIL-STD-286, method 504.1.

4.3.6 Ballistics.-Unless otherwise specified herein, the test shall be performed in accordance with applicable provisions of the Ordnance Proof Manual. A weight of charge shall be calculated to give the prescribed service velocity. A ten round uniformity series shall be fired alternately with ten rounds from a standard reference lot of propellant. The mean max. pressure, mean velocity and the velocity standard deviation for the test and reference rounds shall be calculated separately. If the corrected mean velocity of the test rounds differs by more than one percent from the prescribed service velocity,

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re-evaluations of the established charge and a new uniformity series shall be accomplished prior to determination of the final recommended charge weight.

4.3.7 Air pressure test for containers.

4.3.7.1 One half to one pound test (see dwg. 8858848).-The container shall be subjected to an internal pressure of 1/2 to 1 pound per square inch by a method satisfactory to the contracting officer's representative. A water manometer shall be assembled in the system. A drop of 0.7 inches on the manometer in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level B.-The propellant shall be packed in boxes complying with dwg. 76-4-46 or 76-4-56 (as required). Containers shall be closed, sealed and tested in accordance with the requirements of dwg. 8858848.

5.2 Marking.-The containers shall be marked in accordance with dwg. 8858848.

5.3 Data card.-Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing requirements (see 5.1.1).
c. Details of manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture. Any

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departure from this manufacturing process must be submitted for approval, in writing, to the procuring activity prior to initiation.

6.2 Surveillance sample.-The expense of surveillance testing shall be borne by the Government. All samples should be packed in air-tight containers and marked to show the propellant designation, the projectile for which manufactured, lot number, manufacturer date of sampling, contract number and number of pounds in the lot.

6.3 Standard deviation.-The standard deviation will be calculated using (N-1) as the divisor in a standard statistical technique equivalent to that shown in Standard MIL-STD-414, section B example B1.

6.4 Submission of test data.-In addition to normal distribution of records, when the propellant is procured by the department of the Army, one copy of all ballistic test data including charge establishment firing data, will be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
Attn: ORDBB-ND4

6.5 Inspection Code number.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
Army-Ordnance Corps

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